

04.-Redes-y-Conectividad

En esta actividad aprenderemos a **crear redes** y **gestionar redes personalizadas**. Aplicar los comandos de creación, inspección, renombre y eliminación. Conectar/desconectar contenedores ya creados y también utilizar redes al momento de su creación.

Creación de redes

```
docker network create RED-MOSQUITTO
```

```
docker network create RED-KAFKA
```

```
docker network create RED-GRAFANA
```

```
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network create RED-MOSQUITTO
fdde28a31a204d7c508b649e9f49fece64f9196dd1bc5a845c6508ee55db3cc6
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network create RED-KAFKA
be565d2b29f604db7ca5903692d1a01dc2a8c098cc0638c4050adbe58867ff21
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network create RED-GRAFANA
fee2f93cef58bf12d3c656a20d4d3ae69bfb4e869a297dd0b2a2568dd0a40069
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
fee2f93cef58	RED-GRAFANA	bridge	local
be565d2b29f6	RED-KAFKA	bridge	local
fdde28a31a20	RED-MOSQUITTO	bridge	local
10f57f843433	bridge	bridge	local
7cfb94847053	host	host	local
8d0e766d90b6	none	null	local

Crearemos contenedores y los uniremos a las Redes Creadas

```
docker container run -d \
-p 1883:1883 \
--name DOCKER-CONTAINER-MOSQUITTO \
--network RED-MOSQUITTO \
eclipse-mosquitto:2.0.18
```

```
docker container run -d \
-p 9092:9092 \
--name DOCKER-CONTAINER-KAFKA \
--network RED-KAFKA \
apache/Kafka:3.8.0
```

```
docker container run -d \
-p 3000:3000 \
--name DOCKER-CONTAINER-GRAFANA \
--network RED-GRAFANA \
grafana/grafana:11.3.0
```

```

jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \
> -p 9092:9092 \
> --name DOCKER-CONTAINER-KAFKA \
> --network RED-KAFKA \
> apache/kafka:3.8.0
fc2e83866ab4cbbd18c7755280d2453ca0f5d2935e24e35089fe2bb6b02f36eb
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \
> -p 1883:1883 \
> --name DOCKER-CONTAINER-MOSQUITTO \
> --network RED-MOSQUITTO \
> eclipse-mosquitto:2.0.18
b5d6a52e1cca1448ae64056e30b3f49d3cf84e8b16d6380157438fe6fa526ac1

```

```

jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \
> -p 9092:9092 \
> --name DOCKER-CONTAINER-KAFKA \
> --network RED-KAFKA \
> apache/kafka:3.8.0
fc2e83866ab4cbbd18c7755280d2453ca0f5d2935e24e35089fe2bb6b02f36eb
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker container run -d \
> -p 1883:1883 \
> --name DOCKER-CONTAINER-MOSQUITTO \
> --network RED-MOSQUITTO \
> eclipse-mosquitto:2.0.18
b5d6a52e1cca1448ae64056e30b3f49d3cf84e8b16d6380157438fe6fa526ac1

```

Inspeccionamos las redes creadas

docker network inspect RED-GRAFANA

```

"Labels": {},
"Containers": {
  "99aa574cd8a0384c3b8cff493396edd9274542b1854d9ccfd59c7518b40ab22": {
    "Name": "DOCKER-CONTAINER-GRAFANA",
    "EndpointID": "f607c9ba7fbbc277e50c40373a60be11b1770b0e6b955577dfd23bf06cea5940",
    "MacAddress": "22:cf:3b:77:10:37",
    "IPv4Address": "172.20.0.2/16",
    "IPv6Address": ""
  }
},

```

docker network inspect RED-MOSQUITTO

```

"Labels": {},
"Containers": {
  "b5d6a52e1cca1448ae64056e30b3f49d3cf84e8b16d6380157438fe6fa526ac1": {
    "Name": "DOCKER-CONTAINER-MOSQUITTO",
    "EndpointID": "556264dc141a92b398ba12a9407841050f1c7b672da555b81743d05f1d8cc3a5",
    "MacAddress": "ae:47:ad:c8:ca:ab",
    "IPv4Address": "172.18.0.2/16",
    "IPv6Address": ""
  }
},

```

docker network inspect RED-KAFKA

```

"Labels": {},
"Containers": {
  "fc2e83866ab4cbbd18c7755280d2453ca0f5d2935e24e35089fe2bb6b02f36eb": {
    "Name": "DOCKER-CONTAINER-KAFKA",
    "EndpointID": "3a8260c5d40e9593a8c35bef3f494ad2e2dd9e49ad2bc75ee3e80a2d9229b6ee",
    "MacAddress": "86:08:1f:40:81:52",
    "IPv4Address": "172.19.0.2/16",
    "IPv6Address": ""
  }
},

```

Desconectar los contenedores de las redes creadas

docker network disconnect RED-GRAFANA DOCKER-CONTAINER-GRAFANA

docker network disconnect RED-MOSQUITTO DOCKER-CONTAINER-MOSQUITTO

docker network disconnect RED-KAFKA DOCKER-CONTAINER-KAFKA

```
Run 'docker --help' for more information
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network disconnect RED-GRAFANA DOCKER-CONTAINER-GRAFANA
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network disconnect RED-MOSQUITTO DOCKER-CONTAINER-MOSQUITTO
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network disconnect RED-KAFKA DOCKER-CONTAINER-KAFKA
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network rm RED-GRAFANA
Error response from daemon: network RED-GRAFANA not found
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network rm RED-MOSQUITTO
Error response from daemon: network RED-MOSQUITTO not found
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network rm RED-KAFKA
Error response from daemon: network RED-KAFKA not found
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ exit
exit status 1
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$ docker network rm RED-KAFKA
Error response from daemon: network RED-KAFKA not found
jezuz@DESKTOP-NQ62FUL:/mnt/c/Users/Jesus/Desktop/Galaxy Training/Docker/Examen-Final$
```

Eliminación de redes

docker network rm RED-GRAFANA

docker network rm RED-MOSQUITTO

docker network rm RED-KAFKA